Vivek Raj Rampelli
Data Scientist

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Summary

- Data Scientist with around 4 years of experience leveraging Python, Machine Learning, Amazon Web Services (AWS), SQL, and Data Management to extract insights from complex datasets.
- Experience in Python to manipulate data for data loading and extraction and worked with Python libraries like NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, PyTorch, and NLTK.
- Expert in Python for data manipulation and insightful data visualizations, build interactive dashboards using Tableau and Power BI to communicate complex findings to stakeholders effectively.
- Experience in building production quality and large-scale deployment of applications related to natural language processing (NLP) and machine learning algorithms.

Education

University of North Texas Aug 2022 – May 2024
Master of Science in Information Technology *Denton, TX*

Experience

CitiGroup Jan 2024 – Current
Data Scientist II *Texas*

- Developed a predictive model using XGBoost, Random Forest, and Logistic Regression to identify customers at risk of churning, resulting in a 25% reduction in churn rate through targeted retention strategies.
- Sales Insights Enhancement Using Tableau and MySQL: Developed a Tableau and MySQL-based sales insights project, processing over 500,000 data points with a star schema and ETL, enhancing data-driven decisions.
- Employed K-means clustering to identify distinct customer segments, leading to targeted marketing campaigns that increased customer satisfaction by 15% and boosted sales by 20%.
- Attained a significant 32% improvement in overall model performance by strategically combining Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) in a hybrid architecture for complex data analysis tasks.
- Achieved a 20% increase in F1 score for a binary classification model by fine-tuning hyperparameters and exploring different feature engineering techniques.
- Developed compelling data visualizations using R (ggplot2), Python (SciPy, Matplotlib), and Power BI to communicate insights from complex datasets.

Accenture Dec 2021 – Aug 2022
Data Scientist *India*

- Engineered Decision Tree, Random Forest, and SVM models, enabling targeted marketing and custom PyTorch Data Loaders to handle complex data formats increasing data ingestion efficiency by 40%.
- Improved system reliability by 25% through proactive identification and resolution of anomalies using unsupervised learning methods.
- Leveraged AWS EMR to create highly scalable machine learning pipelines, capable of handling training datasets exceeding terabytes in size and achieving approximately 10% faster model deployment times.
- Developed real-time Power BI dashboards to visualize ML model output, providing key performance indicators (KPIs) like accuracy (95%), confidence (80%), and error rates (5%) for informed decision-making.
- Boosted SQL queries and adjusted resource allocation, improving query performance and reducing overall processing time by 20%, leading to faster data analysis and better decision-making for the organization.

Trion IT Solutions Aug 2019 – Nov 2021
Jr Data Scientist *India*

- Built a natural language processing pipeline to extract key insights from customer feedback, and increase customer satisfaction.
- Implemented sentiment classification using VADER, TextBlob, and NLTK libraries for tokenization, lemmatization, and entity recognition for real-time sentiment scoring.

- Accelerated data exploration by 30% using Python data visualization libraries like Matplotlib and Seaborn, enabling insights into data distributions and patterns.
- Engaged in feature engineering activities, including generating feature intersections, normalizing data, and encoding categorical variables using Scikit-learn's preprocessing tools.
- Generated dashboards using Tableau to track health metrics, provided real-time insights for senior management team members, and improved data-driven decision-making by 30%.

Technical Skills

Language/ IDE's: Python, SQL, Jupyter Notebook, Google Colab.
Machine Learning: Linear, Logistic Regression, Decision Trees, Random Forests, Naive Bayes, SVM, XGBoost, K Means.
Deep Learning: CNN, RNN, LSTM, Natural Language Processing (NLP), Large Language Model (LLM).
Visualizations: Tableau, Power BI, Looker.
Cloud Technology: Azure Data Lake, Azure Data Factory, Azure Databricks.
Packages and Frameworks: Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, Keras, NLTK, spaCy, PyTorch, FastAPI.
Database and Tools: SQL Server, MySQL, PostgreSQL, MongoDB, Cassandra, SPSS, LangChain.

Publication

Healthcare Analytics using Big Data Technique

- Reviewed the application of big data analytics (BDA) in healthcare, including techniques for collecting, managing, and analyzing large datasets from hospital records, medical exams, and IoT devices.
- Examined the benefits of BDA in enhancing patient care, customizing treatments, and reducing healthcare costs.
- Discussed the impact of cloud computing, IoT, and other technical advancements on healthcare data management and analytics.
- Analyzed the challenges of large-scale data analytics in healthcare, including data security, scalability, and integration issues.

Projects

Summarization and evaluation of Patent Documents| *Python, NLP, Machine Learning, ETL, Docker*

- Developed and optimized the T5 (Text-to-Text Transfer Transformer) model for a specific NLP task using Python with Hugging Face Transformers and PyTorch, achieving an F1 score of 92%.
- Executed comprehensive data collection, cleaning, and augmentation on a dataset of 50,000 entries to support model training.
- Conducted hyperparameter tuning and cross-validation, improving the model's accuracy from 85% to 93% over baseline models.
- Deployed the NLP model in a production environment, handling multiple requests per day, and meticulously documented the process to ensure reproducibility and scalability.

Automated Data Processing Pipeline on AWS | *AWS, Data Pipelines, ETL, Python, MySQL*

- Streamlined data processing workflows, reducing manual intervention and errors by 50%.
- Adapted to fluctuating workloads without the need for server management.
- Achieved significant cost savings and improved operational efficiency.

Predictive Modeling of India's GDP Growth Rate | *Time series analysis, python, Data pipelining, tableau, ML*

- Built a predictive model to forecast the annual GDP growth rate of India based on historical economic data, using ML techniques.
- Refined data from credible sources and resolved data gaps/inconsistencies issues to catalyze model training.
- Engineered key features influencing GDP growth, including socio-political factors, external influences, and economic indicators.
- Amplified the model accuracy by defining the relationship among diverse variables through time series analysis and ML algorithms.
- Examined the accuracy of predictive models by calculating metrics like MSE, RMSE, and R-squared.
- Created economic forecasts for 2035, delivering valuable insights to inform decision-making for policymakers and businesses.

Certifications

AWS Cloud Certification

Database operations on MariaDB database by Infosys